





# Scheme of marking

①

n-type

(Wed) 01<sup>st</sup> April, 2026

p-type

(1) - Donor dopants

Acceptor dopants

(1) - majority carriers are electrons

majority carriers are holes

(1) - minority are holes

minority are electrons

(2) - Eg n-Si is doped with pentavalent impurities like P, As, etc

p-Si is doped with trivalent impurities like B, Al, etc

(Optional)  $\sigma \approx ne\mu_e$

$\sigma \approx pe\mu_h$

② (a) Case  $\lambda$

(1) (i) K.E. =  $E = \frac{h^2}{2m\lambda^2}$

(1) (ii) Pot diff =  $V = \frac{h^2}{2m\lambda^2}$

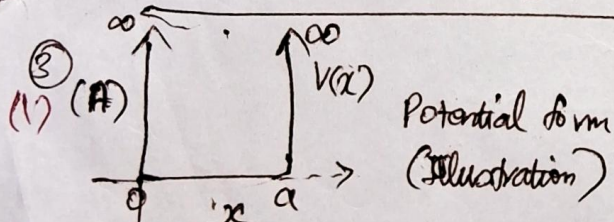
(1) (iii) Temperature  $T = \frac{h^2}{3mk_B\lambda^2}$

(b) (i)  $E = 10 \text{ KeV}$

$$\lambda = \frac{h}{\sqrt{2mE}} = \frac{6.625 \cdot 10^{-34}}{\sqrt{2 \cdot 9.1 \cdot 10^{-31} \cdot 10^4 \cdot 1.6}} = \frac{6.625 \cdot 10^{-34}}{\sqrt{2.912 \cdot 10^{-25}}} = 1.25 \cdot 10^{-11} \text{ m} = 0.125 \text{ \AA}$$

(1) (ii)  $V = 100 \text{ V}$

$$\lambda = \frac{h}{\sqrt{2mV}} = \frac{6.625 \cdot 10^{-34}}{\sqrt{2 \cdot 9.1 \cdot 10^{-31} \cdot 100}} = 1.25 \cdot 10^{-10} \text{ m} = 1.25 \text{ \AA}$$



(1) (B) Trial solution  $\Psi(x) = A \sin Kx + B \cos Kx$

(1) (E) Normality condition

$$\int_{-\infty}^{\infty} |\Psi|^2 dx = 1 \Leftrightarrow \int_0^a B^2 \sin^2 \frac{n\pi x}{a} dx = 1 \Leftrightarrow B = \sqrt{\frac{2}{a}}$$

(1) (C) Left Boundary condition

$$\Psi(x=0) = 0 \Leftrightarrow B = 0$$

(1) (F) Eigenfunctions

$$\Psi_n(x) = \sqrt{\frac{2}{a}} \sin \frac{n\pi}{a} x, n \in \mathbb{Z}^+$$

(1) (D) Right Boundary condition

$$\Psi(x=a) = 0 \Leftrightarrow A \sin Ka = 0$$

$$\Leftrightarrow \sin Ka = 0$$

$$\Leftrightarrow Ka = n\pi, n \in \mathbb{Z}^+$$

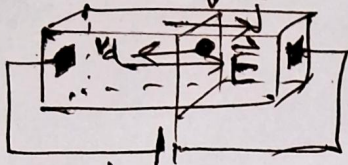
$$\Leftrightarrow K = \frac{n\pi}{a}, n \in \mathbb{Z}^+$$

(1) (G) Eigenvalues

$$E_n = \frac{\hbar^2 K^2}{2m} = \frac{\hbar^2 n^2 \pi^2}{2ma^2} = \frac{n^2 \hbar^2}{8ma^2}$$



④ (a) (i) Current Drift motion illustration



$$\left. \begin{aligned} a &= \frac{eE}{m} \\ v_d &= \frac{eE\tau}{m} \end{aligned} \right\} (2) \quad \left. \begin{aligned} j &= \sigma E \\ \Rightarrow \sigma &= \frac{ne^2\tau}{m} \end{aligned} \right\} (2)$$

$$\left. \begin{aligned} j &= nev_d \\ j &= \frac{ne^2\tau}{m} E \end{aligned} \right\} (2)$$

④ (b)  $\rho = 1.54 \cdot 10^{-8} \Omega m$   
 $E = 2 V m^{-1}$   
 $n = 5.8 \cdot 10^{28} m^{-3}$

and  $v_d = \frac{e\tau}{m} E$   
 $= \frac{1.6 \cdot 10^{-19} \cdot 4 \cdot 10^{-4} \cdot 2}{9.1 \cdot 10^{-31}}$

$$\frac{1}{\rho} = \sigma = \frac{ne^2\tau}{m} \Rightarrow \tau = \frac{m}{\rho ne^2} \approx 1.28 \cdot 10^{-33+31} = 1.28 \cdot 10^{-2} ms^{-1}$$

③ (b)  $L = 0.18 nm = 1.8 \cdot 10^{-10} m$

$$\begin{aligned} &= \frac{9.1 \cdot 10^{-31}}{1.54 \cdot 10^{-8} \cdot 5.8 \cdot 10^{28} \cdot 1.6 \cdot 1.6 \cdot 10^{-38}} \\ &= \frac{1}{10^{-31+18}} \\ &= \frac{2.4}{4 \cdot 10^{-14}} s \end{aligned}$$

2nd excited state

$$\Rightarrow E_n = \frac{n^2 h^2}{8mL^2} \quad \left[ \tau \approx 40 fs \right] \quad (2)$$

$$= \frac{9 \cdot 6.625 \cdot 6.625 \cdot 10^{-34} \cdot 10^{-34}}{8 \cdot 9.1 \cdot 10^{-31} \cdot 1.8 \cdot 1.8 \cdot 10^{-20}}$$

$$= \frac{1.8}{1.6 \cdot 10^{-19}} eV \approx 1.1 \cdot 10^{-17+19} = 100 eV \quad (2)$$